

Lucas Root

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Sophomore Computer Science and Cybersecurity Engineering student in the Honors Program at the University of Kansas seeking a software engineering internship.

EDUCATION

University of Kansas <i>B.S. in Computer Science & Cybersecurity Engineering, Honors College</i> – 4.0 GPA — Coursework: Programming 2, Software Engineering, Discrete Structures, Embedded Systems	Lawrence, KS <i>Expected May 2028</i>
Menlo-Atherton High School <i>High School Diploma, 4.08 GPA</i>	Menlo Park, CA <i>May 2024</i>

PROJECTS

TURT: Custom Unix Shell 🛠️ <i>C, Linux System Calls, Makefile</i> – Engineered a POSIX-compliant shell in C, managing process execution via <code>fork</code> , <code>execvp</code> , and <code>wait</code> . – Implemented custom tokenizer for argument parsing and dynamic memory allocation (verified via Valgrind). – Designed signal handling interrupts and built-in commands to replicate core Bash functionality.	Oct. 2025 - Present
Praxis: Student Social Network 🛠️ <i>Rust, Next.js, Postgres</i> – Architecting a full-stack platform connecting students for project collaboration. – Developing a high-performance backend in Rust to ensure type safety and low-latency database queries. – Designing a relational database schema in PostgreSQL to manage user relationships and project feeds.	Jan. 2026 - Present
Rust CLI Note Manager 🛠️ <i>Rust, Serde</i> – Built a terminal-based CRUD application to master Rust's ownership model and borrow checker. – Implemented custom data serialization traits using Serde for efficient local file storage.	Nov. 2025 - Present
Syllabus2Cal 🛠️ <i>TypeScript, Next.js, OpenAI API</i> – Built a tool converting unstructured syllabus text into structured .ics data using LLM function calling. – Optimized API usage costs and response times through effective prompt engineering and caching.	Jan. 2025 - Present

EXPERIENCE

Navy Midshipman <i>US Navy ROTC, University of Kansas</i> – Completed New Student Indoctrination and Orientation; recognized for leadership potential.	Aug. 2024 – Dec. 2024 <i>Lawrence, KS</i>
Waiter / Busboy <i>Dessert Republic</i> – Provided customer service in a bilingual (English/Cantonese) environment at a family-owned restaurant.	Jun. 2022 – Nov. 2022 <i>San Mateo, CA</i>

RELEVANT EXPERIENCE

Capture The Flag Challenges <i>OverTheWire, VICEROY ShieldBearer & Sentinel Watch</i> – Active participant in cybersecurity challenges focusing on web exploitation, network reconnaissance, and Linux privilege escalation. – VICEROY ShieldBearer: placed 9th of 44 — VICEROY Sentinel Watch: placed 8th – Did challenges on OverTheWire to learn the basics of Unix/Linux commands.	Dec. 2024 - Present
C++ Linear Algebra Library <i>Academic Lab</i> – Implemented a high-performance Matrix class in C++ handling operations like multiplication, transposition, and inversion. – Optimized memory usage for large datasets using pointers and manual resource management.	Fall 2025

LEADERSHIP & AWARDS

President, Kansas Crew <i>University of Kansas</i> – Lead logistics, recruitment, and community partnerships for the official university rowing club.	Aug. 2024 – Present <i>Lawrence, KS</i>
Varsity Rower, NorCal Crew <i>Redwood City, CA</i> – Competed nationally (13th at 2023 Head of the Charles); coached novice rowers ages 11–15.	Aug. 2019 – May 2024
KU Excellence Scholarship <i>University of Kansas</i>	2024

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, Java, C#, JavaScript/TypeScript, SQL
Developer Tools: Git/GitHub, Docker, Linux/Unix, GDB, Valgrind, Make, VSCode
Concepts: Memory Management, OS Architecture, Embedded Systems, Data Structures